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Conjugada armónica

Teorema 1 (Conjugada armónica). *Sea* $f = u + iv \in \mathcal{H}(\Omega)$ *con* $u, v \in \mathcal{C}^2(\Omega)$

$$\implies u, v \text{ son funciones armónicas en } \Omega.$$

En este caso, v *es la* ***conjugada armónica*** *de* u.

***Demostración*:** Por las ecuaciones de Cauchy-Riemann, tenemos que

$$u_{xx} = (u_x)_x = (v_y)_x \wedge u_{yy} = (u_y)_y = (-v_x)_y \underset{\substack{\uparrow \\ v \in \mathcal{C}^2(\Omega)}}{\implies} u_{xx} = -u_{yy}.$$

Análogamente, $v_{xx} = -v_{yy}$, luego u, v son armónicas. ∎

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